



BITS Pilani
Pilani Campus

API-driven Cloud Native Solutions

Dr. Shreyas Rao
Associate Prof. (Off Campus), CSIS, BITS-Pilani



CC ZG506

Lecture No. 13

Contact Session wise Coverage

Session	Coverage	
CS01	API Basics (OpenAPI, REST, gRPC, GraphQL)	} Basics
CS02	API Basics + Cloud Native Application	
CS03	Cloud Native Application	
CS04	Introduction to Data Science & Machine Learning, DataOps & MLOps	} Data Science & ML
CS05	API driven Cloud-native Data Pipeline	
CS06	API-driven ML pipelines	
CS07	Tools review	} Cognitive AI
CS08	Basic APIs for AI Applications - NLP, CV, Speech recognition	
CS09	Language Models (LLMs)	
CS10	Fine Tuning + RAG Models	
CS11	RAG Models	} IoT
CS12	Cloud Native Application Deployment	
CS13	IoT and Data Analytics	
CS14	APIs for IoT and Data Analytics	
CS15	Serverless Functions; Case study	
CS16	Course review and discussion	

Agenda

Data Analytics

- Overview of Data Analytics
- Types of Data Analytics



BITS Pilani
Pilani Campus



Data Analytics

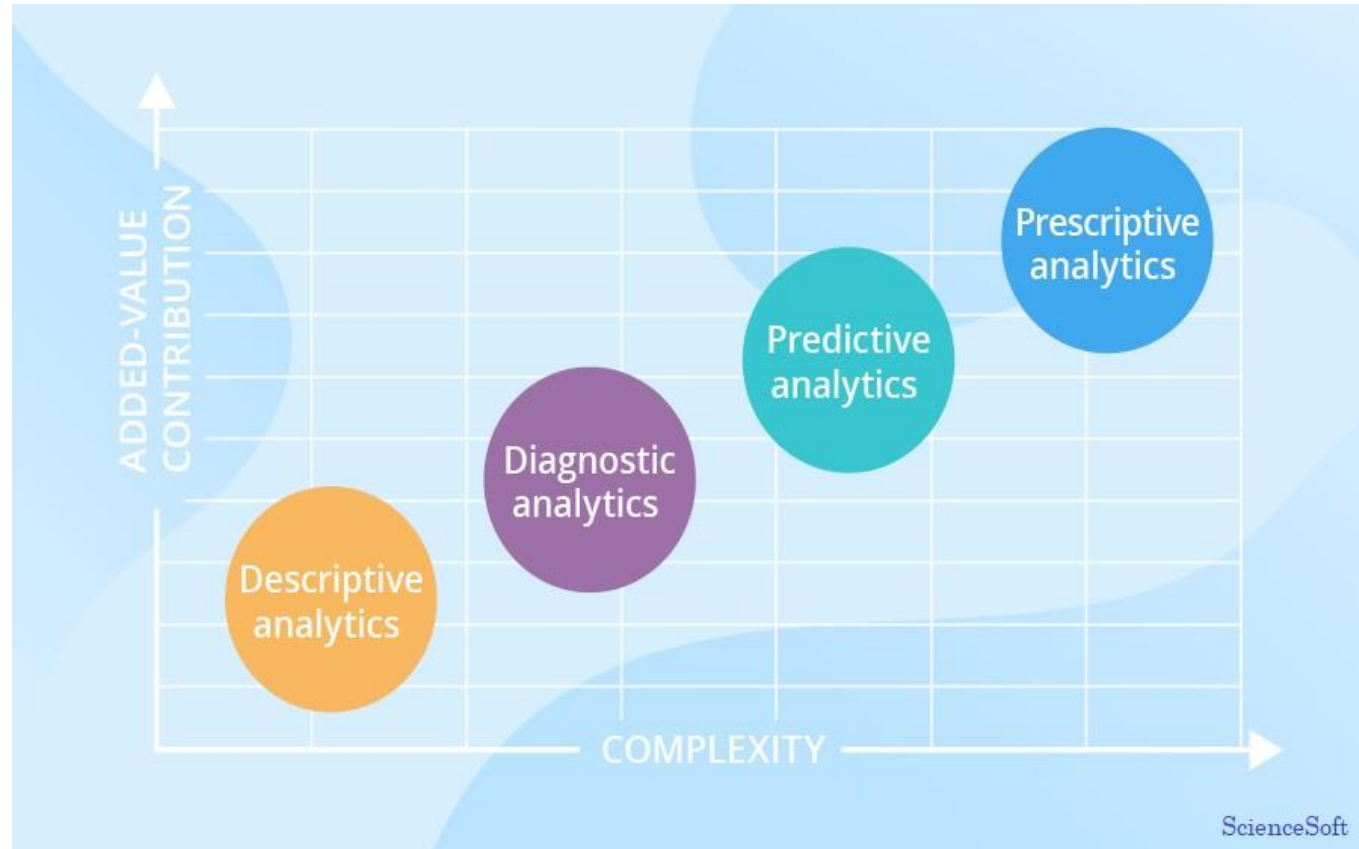
Data Analytics

Data analytics is defined as a process of cleaning, transforming, and modeling data to discover useful information for business decision-making.

4 different types of analytics

1. Descriptive Analytics
2. Diagnostic Analytics
3. Predictive Analytics
4. Prescriptive Analytics

Data Analytics



Goals of Data Analytics



To predict something

- whether a transaction is a fraud or not [Banking]
- whether it will rain on a particular day [Weather Forecast]
- whether a tumor is benign or malignant [Cancer Prediction, Healthcare]

To find patterns in the data

- finding the top 10 coldest days in the year [Weather Forecast]
- which pages are visited the most on a particular website [Web Traffic Rank]
- finding the most searched celebrity in a particular year [Awards]

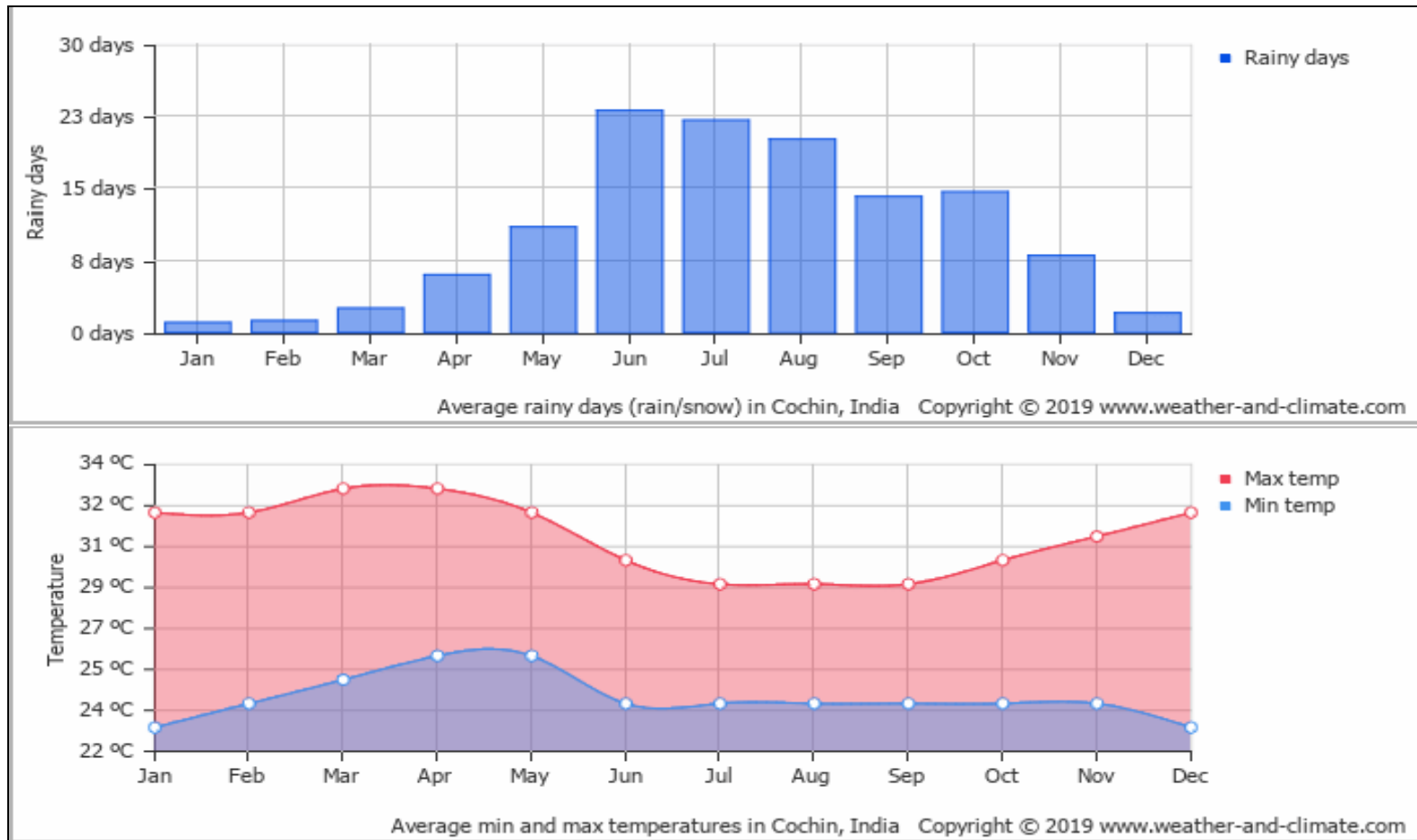
To find relationships in the data

- finding similar news articles [Bing, Google]
- finding similar patients in an electronic health record system [Healthcare]
- finding related products on an e-commerce website [Recommendation]
- finding correlation between news items and stock prices

Descriptive Analytics

- Answers the question of **what happened**.
- Summarize past data usually in the form of dashboards.
- **Insights into the past**.
- Also known as **statistical analysis**.
- Raw data from multiple data sources.

Descriptive Analytics Example-I



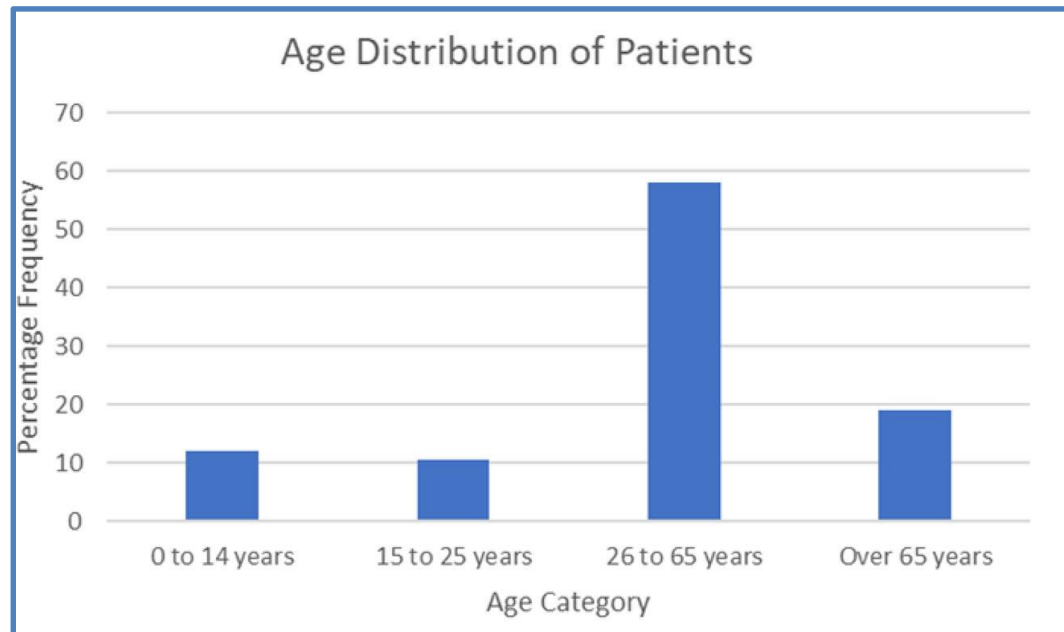
Descriptive Analytics Example-II

Table 1. Case characteristics.

Type of case	(n)	%
New case	216	47.37
Follow-up / Review	212	46.49
Report Consultation	28	6.14
Total number of cases	456	100

Table 2. Specialization/Department categories.

Specialization categories	(n)	%
Basic	203	44.7
Dermatology	29	6.4
General surgery	10	2.2
Medicine	74	16.2
Obstetrics and gynecology	32	7
Orthopedics	24	5.3
Pediatrics	34	7.5
Secondary	55	12
Gastroenterology	22	4.8
Ophthalmology	12	2.6
Psychiatry	21	4.6
Tertiary	198	43.3
Cardiology	23	5
Cardiothoracic surgery	6	1.3
Endocrinology	32	7
Infectious diseases	7	1.5
Medical oncology	5	1.1
Nephrology	26	5.7
Neuro surgery	14	3.1
Neurology	33	7.2
Nuclear medicine	23	5
OTO rhino laryngology	5	1.1
Pediatric hematology and oncology	1	0.2
Pediatric surgery	2	0.4
Pulmonary medicine	9	2
Radiotherapy	2	0.4
Surgical gastroenterology	1	0.2
Surgical oncology	2	0.4
Urology	7	1.5
Grand total	456	100.0



Descriptive Analytics Techniques

Techniques

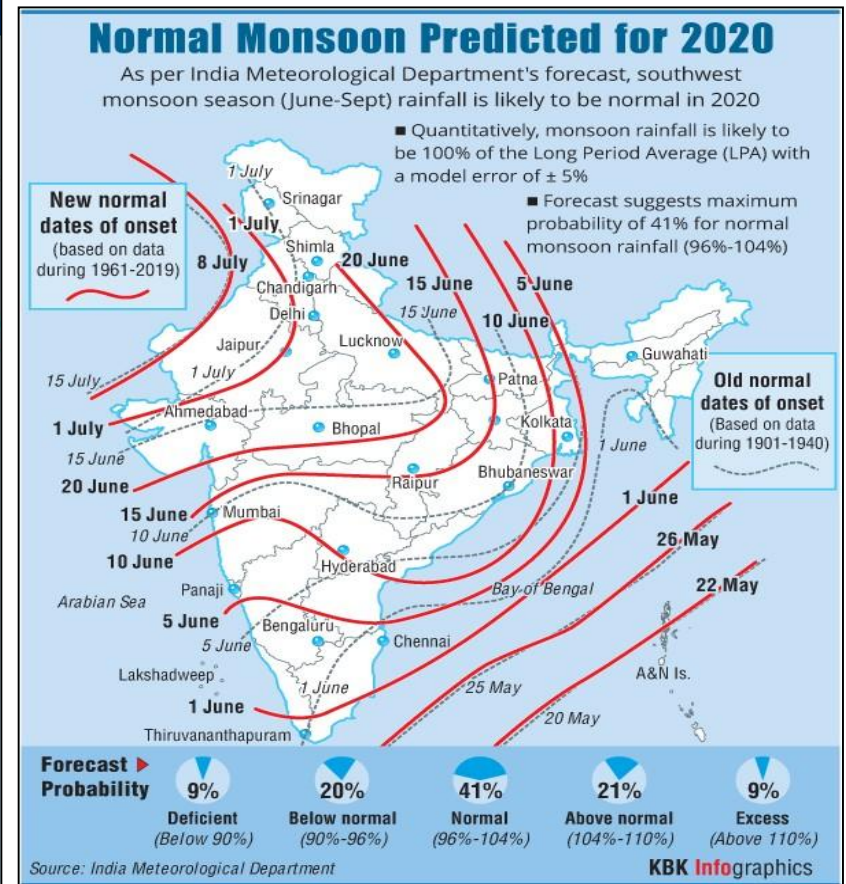
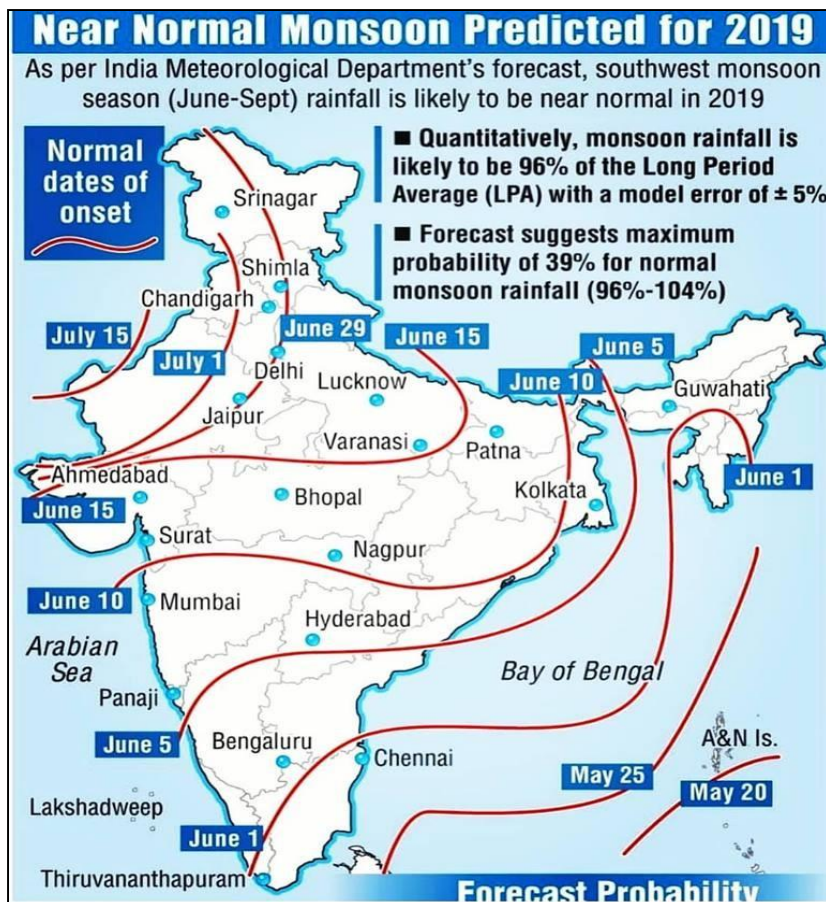
- Descriptive Statistics - histogram, correlation
- Data Visualization
- Exploratory Analysis [Seaborn Library in Python]

Diagnostic Analytics

- Answers the question of **why something happened**.
- Gives in-depth insights into data.
- **Identify relationship** between data and **identify patterns** of behavior.
- Diagnostic analytics is a form of data analytics that builds on descriptive analytics to help you understand why something happened in the past.
- Often, diagnostic analysis is referred to as **root cause analysis**. It involves processes such as data discovery, data mining, and drill down and drill through.

Diagnostic Analytics Example

What is the effect of global warming in the Southwest monsoon?



Diagnostic Analytics Techniques

- Pattern recognition to identify patterns.
- Linear / Logistic regression to identify relationship.
- Neural Network
- Deep Learning techniques

Predictive Analytics

- Answers the question of **what is likely to happen**.
- **Predict future trends**.
- Being able to predict allows one to make better decisions.
- Analysis based on machine or deep learning.
- Accuracy of the forecasting or prediction highly depends on data quality and stability of the situation.

Predictive Analytics Techniques

Techniques / Algorithms:

- Regression
- Classification
- ML algorithms like Linear regression, Logistic regression, SVM
- Deep Learning techniques



Predictive Analytics Example

Covid Patient Discharge Prediction (Dataset: 2nd Wave April-2021 to June 2021)

Type of Project: Machine Learning

Dataset size: 1233 patients suffering from Covid

Variables:

X: Age, Gender, Co_morbid, Admit Date, Discharge date, days of stay, covid_severity

Y: Discharge Type (Recovered, Expired)

Exploratory Data Analysis: Univariate, Bivariate, Multivariate

Models applied: Support Vector Machine, Naïve Bayes, Logistic Regression, Decision Trees, KNN, ANN, Random Forest

Best Accuracy: Random Forest (92%)



Prescriptive Analytics

- Answers the question of **what might happen**.
- **Data-driven decision making and corrective actions**
- Prescribe what action to take to eliminate a future problem or take full advantage of a promising trend.
- Need **historical internal data** and **external information like trends**.
- Analysis based on machine or deep learning, business rules.
- Use of AI to improve decision making.

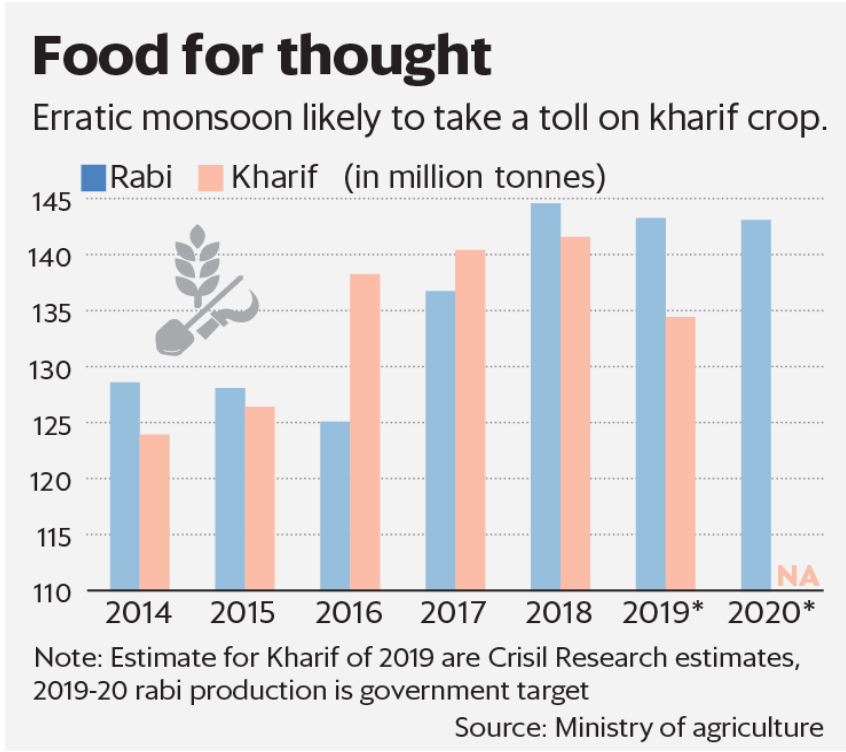
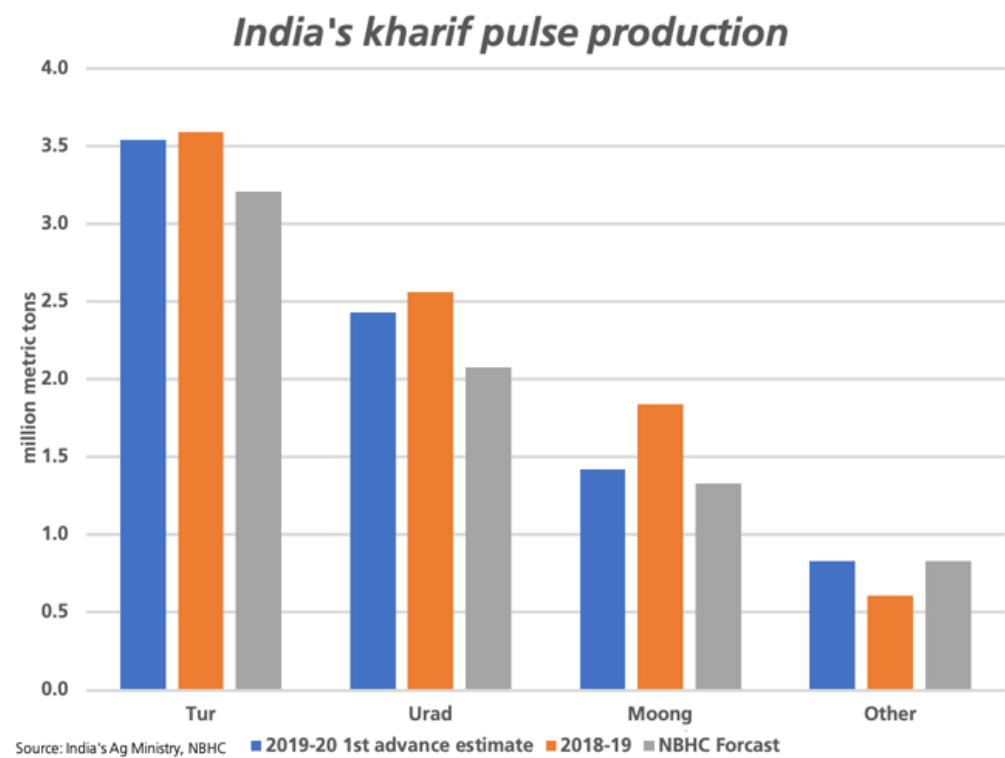


Prescriptive Analytics Example - I

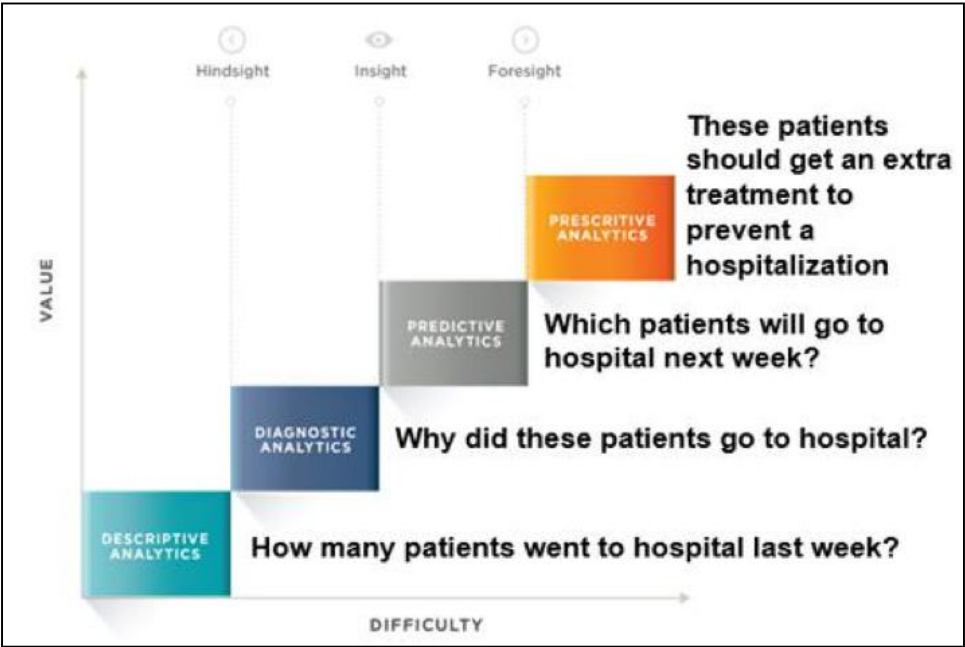
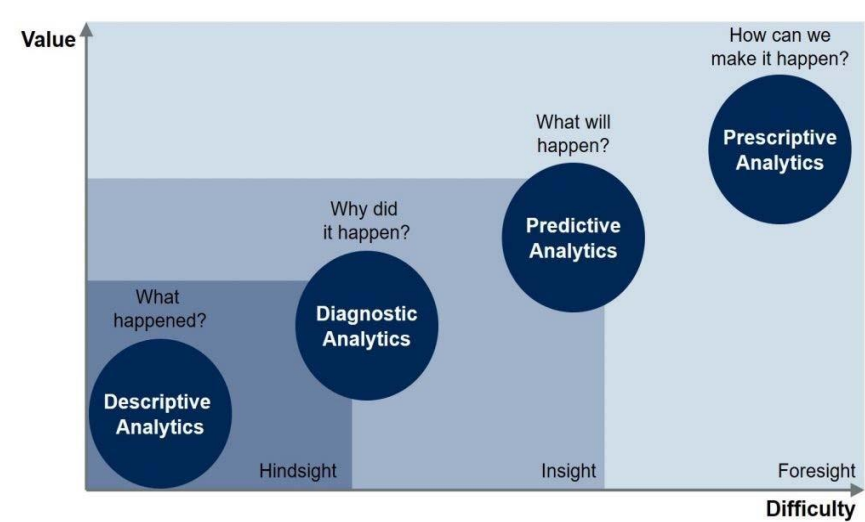
- Apollo Hospitals uses an AI tool to predict the risk of cardiovascular disease.
- The Apollo AI-powered “Cardiovascular Disease Risk” tool will help healthcare providers to predict the risk of cardiac disease in their patients [Predictive Analytics]
- The prediction **initiates intervention early** enough to make a real difference. [Prescriptive]
- The cardiac risk scoring tool is remarkable for the speed in processing data and its accuracy at predicting the probability of a patient developing coronary disease.
- Using the tool, physicians will be enabled to deliver **proactive, pre-emptive and preventive care** for at-risk individuals, improving lives, while mitigating future pressure on healthcare systems.

Prescriptive Analytics Example - II

How can we improve the crop production?



Types of Data Analytics



Types of Data Analytics



Exercise

Instagram Reels allows users to create fun videos and share with their contacts. Users can record 15 second multi-clip videos with audio and effects. Some features include: exploring reels based on subject; following, commenting and liking a reel; identifying trends to create new reels. The reels are released in two versions - public (free for all), and premium (subscription basis).

- Discuss the four analytical tasks that can be performed with respect to the Instagram Reels? [Descriptive, Diagnostic, Predictive and Prescriptive]

Types of Data Analytics



Instagram Reels

Descriptive - How many followers do you have, how many views, comments, likes for your video [free], audience breakdown by country, follower activity per hour [premium]

Diagnostic - Why your video's engagement rate is less. [premium users]

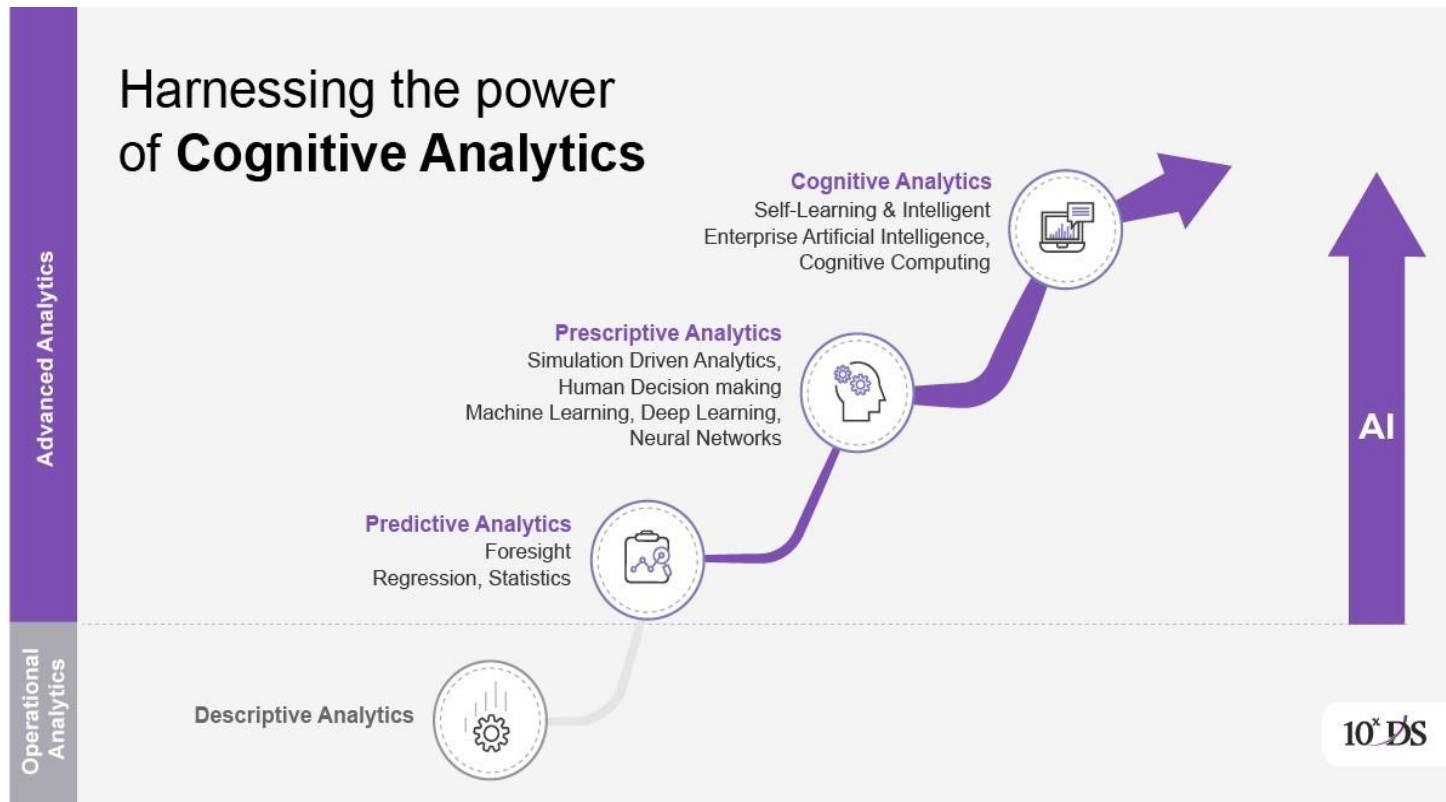
Predictive - Trending topics for you to make video on - their approximate engagement rates [premium]

Prescriptive - Tips to increase average watch time of your videos [premium]

Cognitive Analytics



Cognitive Analytics - What I Don't Know?



Cognitive Analytics



- Next level of Analytics
- Human cognition is based on the **context** and **reasoning**.
- Cognitive systems mimic how humans reason and process.
- Cognitive systems analyze information and draw inferences using probability.
- They continuously learn from data and reprogram themselves.

Definition according to one source:

"The essential distinction between cognitive platforms and artificial intelligence systems is that you want an AI to do something for you. A cognitive platform is something you turn to for collaboration or for advice."

Cognitive Analytics

innovate

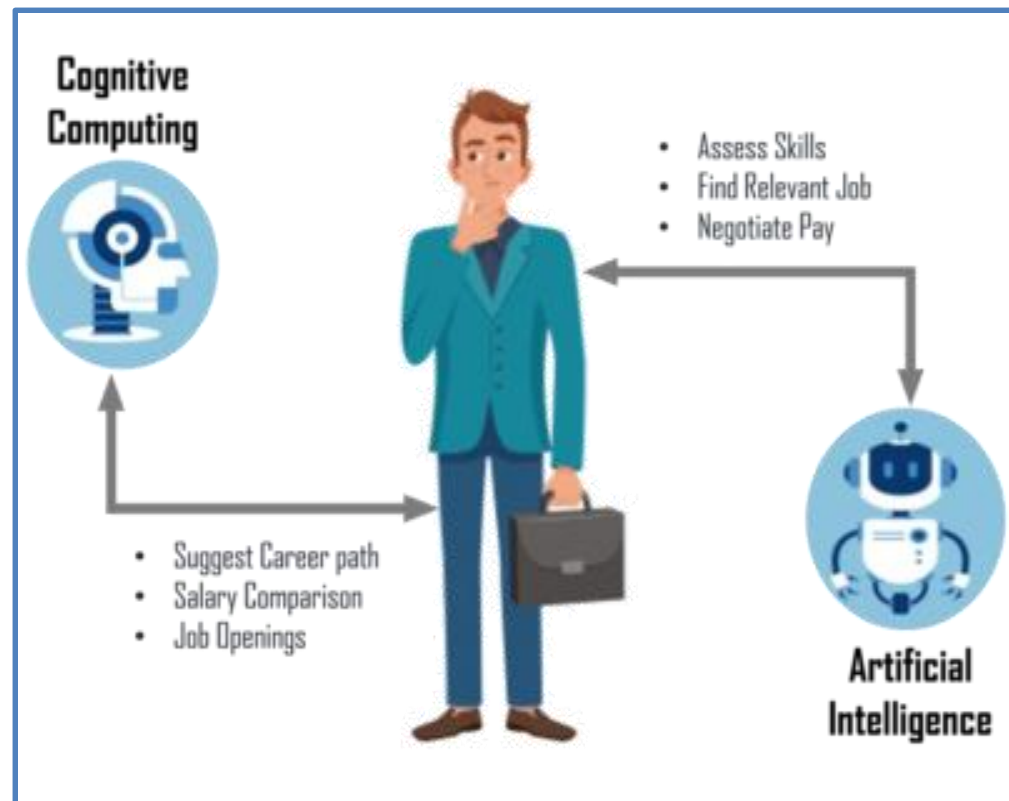
achieve

lead

- Involves Semantics, AI, Machine learning, Deep Learning, Natural Language Processing, and Neural Networks.
- Simulates human thought process to learn from the data and extract the hidden patterns from data.
- Uses all types of data: audio, video, text, images in the analytics process.
- Although this is the top tier of analytics maturity, Cognitive Analytics can be used in the prior levels.

According to Jean Francois Puget:

- "It extends the analytics journey to areas that were unreachable with more classical analytics techniques like business intelligence, statistics, and operations research."



Example



Woebot Mental Health App

- Provides mental health support, using Cognitive Behavioral Therapy (CBT)
- NLP based self-learning App that advises / chats with users on mental health, developed by Stanford University

Category: Conversational AI for Therapy: Woebot applies principles from cognitive analytics and predictive models to shape responses, but its primary function is to provide therapeutic support through guided conversation.

Benefits:

- Using Woebot led to significant reductions in anxiety and depression among people aged 18-28 years old, compared to an information-only control group.
- 85% of participants used Woebot on a daily or almost daily basis.

Cognitive Analytics



Healthcare Diagnosis and Treatment Recommendation

- **How it's used:** Cognitive analytics systems like IBM Watson Health analyze patient records, genetic information, research papers, and clinical trials to suggest personalized treatment options.
- **Example:** Helping doctors detect diseases like cancer early by identifying subtle patterns in MRI scans and medical history.

Cognitive Analytics



Customer Service Chatbots and Virtual Assistants

- **How it's used:** Cognitive systems power chatbots that can understand customer emotions and context to respond more intelligently.
- **Example:** Virtual agents like Amelia or IBM Watson Assistant resolve queries, escalate complex cases, and even offer personalized product recommendations.

Cognitive Analytics



Predictive Maintenance in Manufacturing

- **How it's used:** Cognitive systems analyze sensor data from machines to predict equipment failures before they happen.
- **Example:** GE and Siemens use cognitive analytics to reduce downtime and optimize maintenance schedules in industries like aviation and power generation.



Thank You!